# Introduction to the t-test

**Topic 8 · Thursday, February 4 and Tuesday, February 9**

EDUC 15: Statistics for Education Research
UCI School of Education

# Where are we?

**Description**

Summarize data

Visualize data

**Prediction**

**We are here**

**Explanation**

# The four steps stay. Step 3 changes

Step 1: State a hypothesis about the population: $H_0$

Step 2: Set the criteria: choose $\alpha$ and find the critical region

**Step 3: Observe a sample and compute a test statistic**

Step 4: Reject or maintain $H_0$

# Plan for today

**One tail or two?**

**When we don't know σ**

**The t-test**

**On Tuesday: how big is the effect, and how sure are we?**

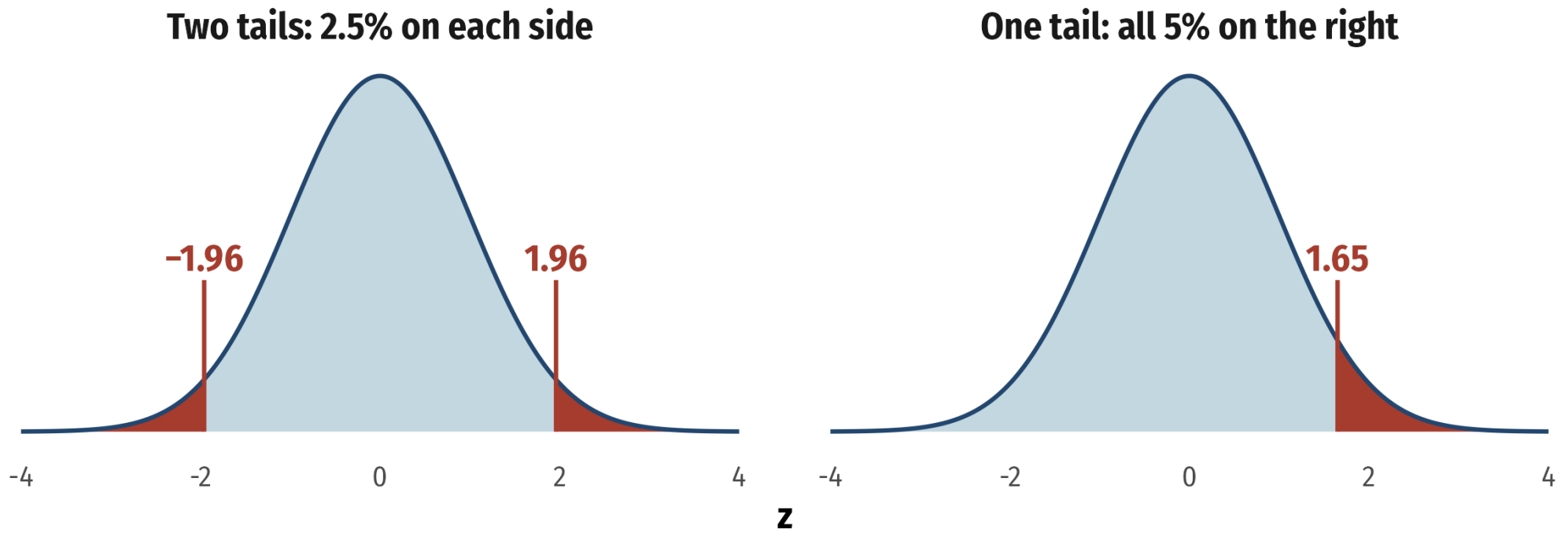
# One tail or two?

# A directional hypothesis picks a side

| | **Non-directional $H_0$** | **Directional $H_0$** |
|---|---|---|
| In symbols | $\mu = 100$ | $\mu \leq 100$ |
| Laptops in class | Laptops don't affect learning | Laptops don't improve learning |
| Cat videos | Cat videos don't change your happiness | Cat videos don't make you happier |

**A directional test says which way the mean should move, not only that it moves**

# Same α, one tail: the cutoff moves in




With α = 0.05 in one tail, the critical z is 1.65, not 1.96. Type I errors stay at 5%

# Predict it: One tail, and a real effect

**$H_0$ says $\mu = 100$, with $\sigma = 15$ and $n = 10$. In truth, $\mu = 113.4$, so $d = 0.89$. A two-tailed test has 80% power**

**We switch to a one-tailed test on the right, and keep $\alpha = 0.05$. What happens to power?**

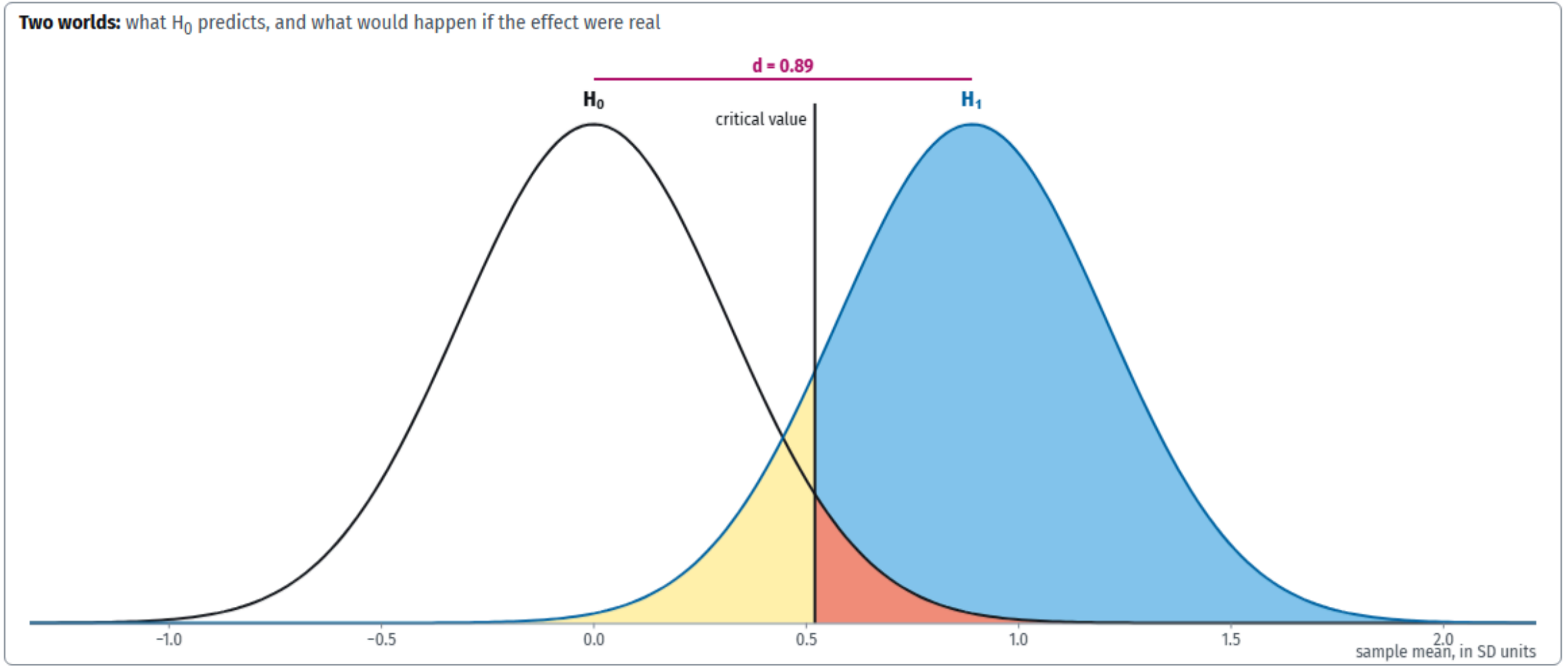
**A** It goes up

**B** It goes down

**C** It stays the same

**D** We can't say without the data

# One tail adds power on that side




Simulator after Kristoffer Magnusson, rpsychologist.com.

# So why not always use one tail?

A one-tailed test is blind on the other side. If the program hurts students, it can't say so

You must pick the side before you see the data. Picking it afterward is really a test at $\alpha = 0.10$

**Two tails is our default from now on, unless a problem says otherwise**

# When we don't know σ

# So far, z has needed σ

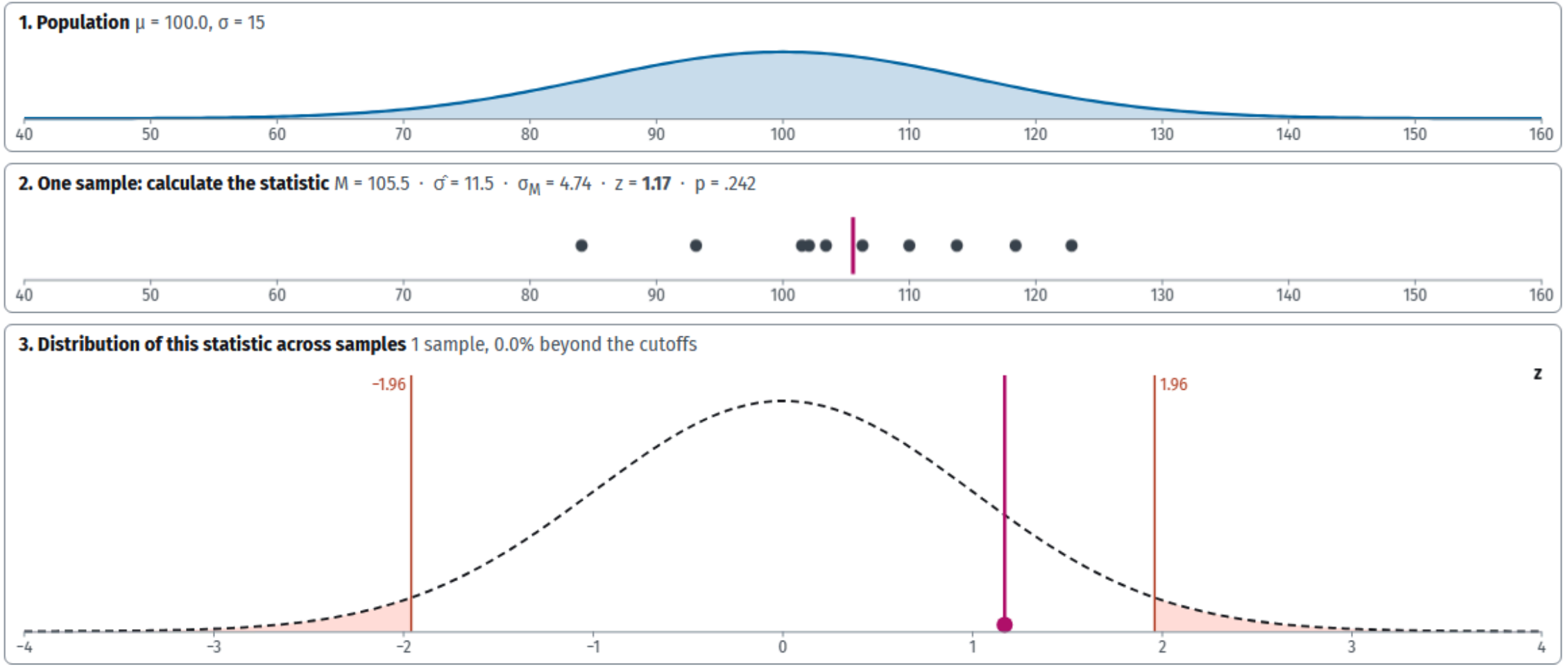
$$z = \frac{M - \mu}{\sigma_M} \qquad \text{where} \qquad \sigma_M = \frac{\sigma}{\sqrt{n}}$$

Top: how far M is from the mean under $H_0$

Bottom: how far a mean typically falls

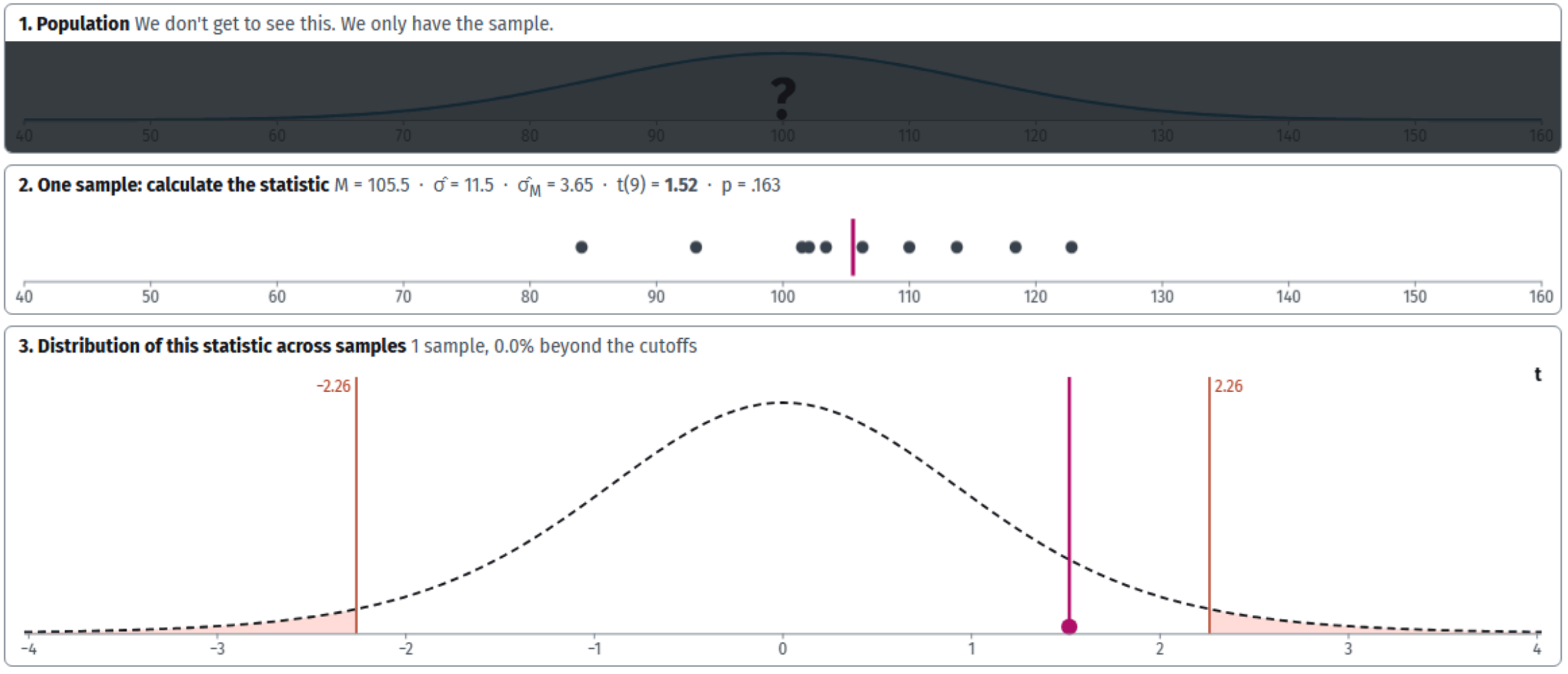
**All quarter, someone has told us σ = 15. Who told us?**

# Here's one sample of 10, with σ known




Simulator after Kristoffer Magnusson, rpsychologist.com.

# In real research, panel 1 goes dark




Simulator after Kristoffer Magnusson, rpsychologist.com.

# So $\hat{\sigma}$ takes the place of $\sigma$

$$z = \frac{M - \mu}{\sigma_M} \qquad \text{becomes} \qquad t = \frac{M - \mu}{\hat{\sigma}_M}$$

**Only the bottom changes: $\hat{\sigma}_M = \hat{\sigma} / \sqrt{n}$. The hat says it's an estimate**

**The new name warns us: this score behaves a little differently**

In the book: $\hat{\sigma}$ is written s, and $\hat{\sigma}_M$ is written $s_M$

# Dividing by n – 1 fixes the variance

| Samples of 5 from our 619 students | Average across samples |
|---|---|
| The sample's own variance, SS / n | 12.13 |
| Our estimate, $\hat{\sigma}^2$ = SS / (n – 1) | 15.17 |
| The truth, $\sigma^2$ | 15.22 |

**SS / n tends to underestimate $\sigma^2$. Dividing by n – 1 gives an unbiased estimate of the variance**

**Then $\hat{\sigma}$ is its square root. From here on, n – 1 is the degrees of freedom: df**

Data: Muralidharan, Singh, and Ganimian (2019). 100,000 samples.

# Predict it: Does swapping in $\hat{\sigma}$ matter?

Respond at **PollEv.com/adb**

**Treat our 619 students as a population. Draw samples of 5, and use $\hat{\sigma}$ in place of $\sigma$. With the true $\sigma$, 5% of samples land beyond $\pm 1.96$. With $\hat{\sigma}$?**

**A** Much less than 5%

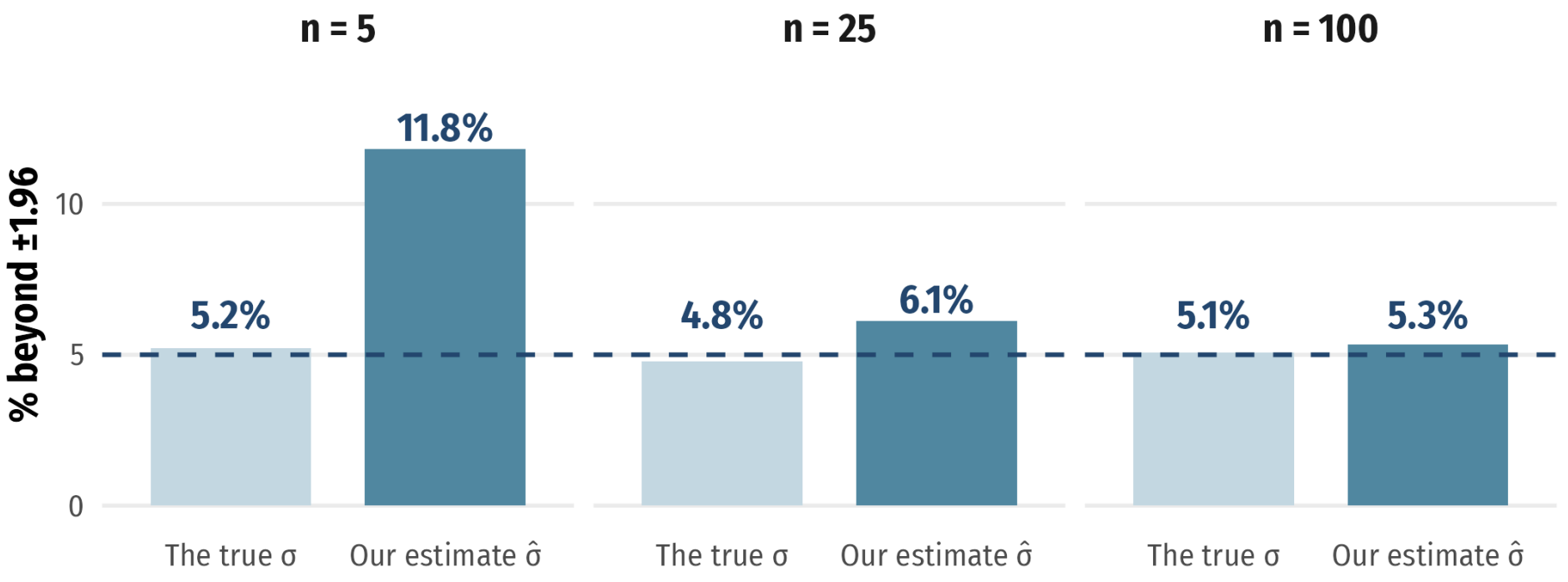
**B** About 5%, the same

**C** More than 5%

**D** It depends on $\mu$

Data: Muralidharan, Singh, and Ganimian (2019).

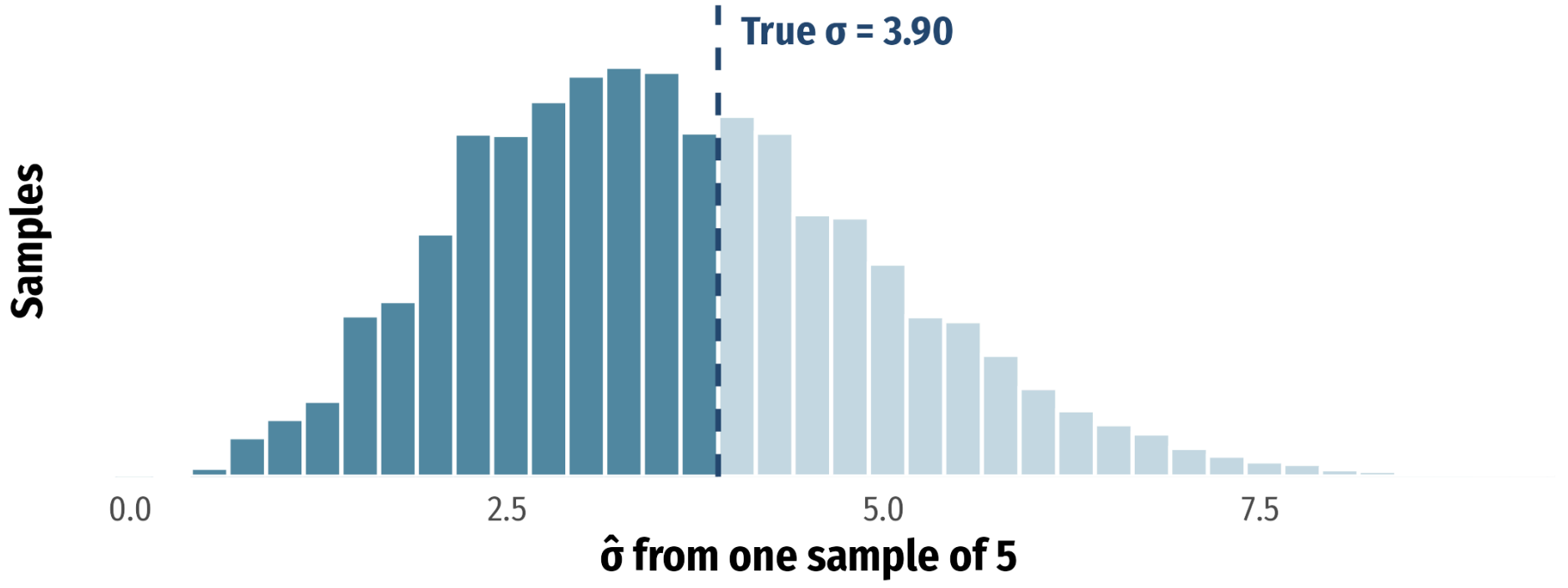
# Small samples bring too many surprises




**As n grows, $\hat{\sigma}$ gets closer to $\sigma$, and the problem fades**

Data: Muralidharan, Singh, and Ganimian (2019). 100,000 samples of each size.

# Why? $\hat{\sigma}$ bounces, and often lands low

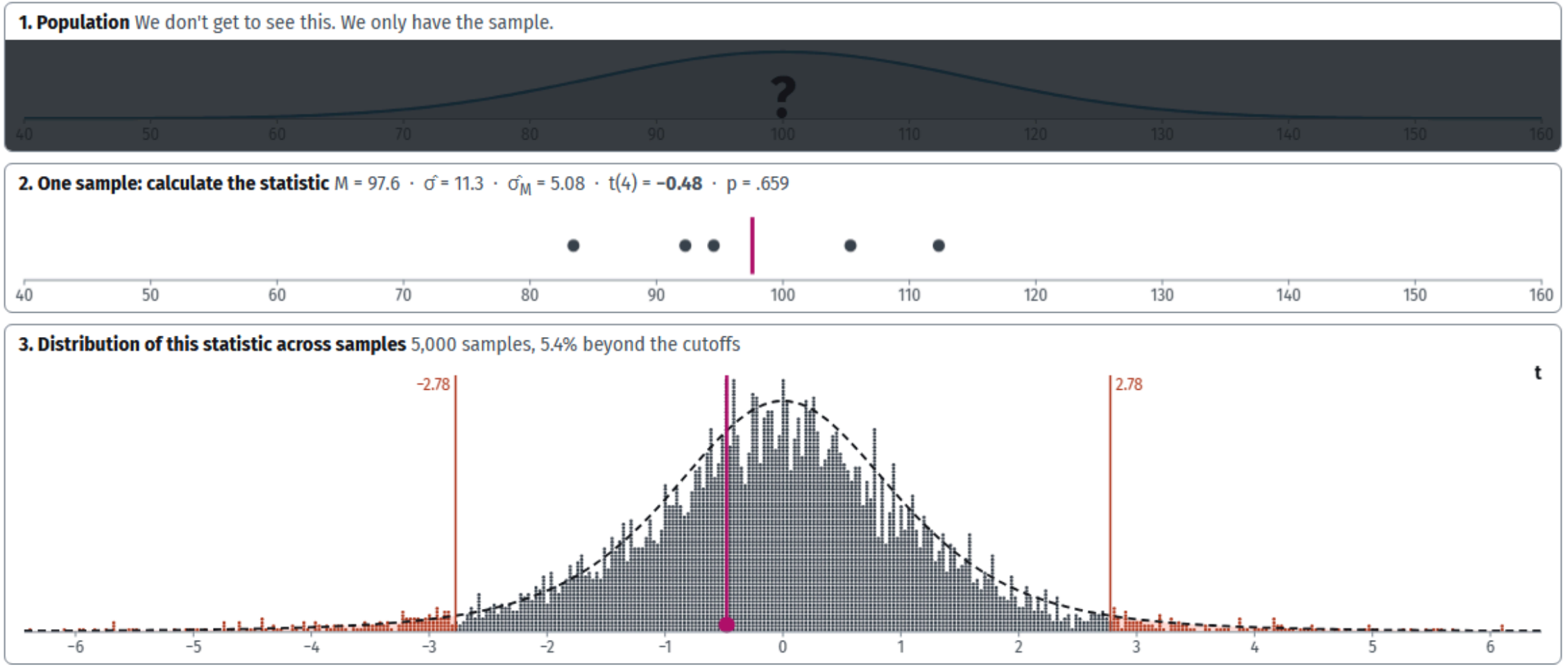



60% of samples give a $\hat{\sigma}$ below $\sigma$. Then the bottom shrinks and the score grows

Data: Muralidharan, Singh, and Ganimian (2019).

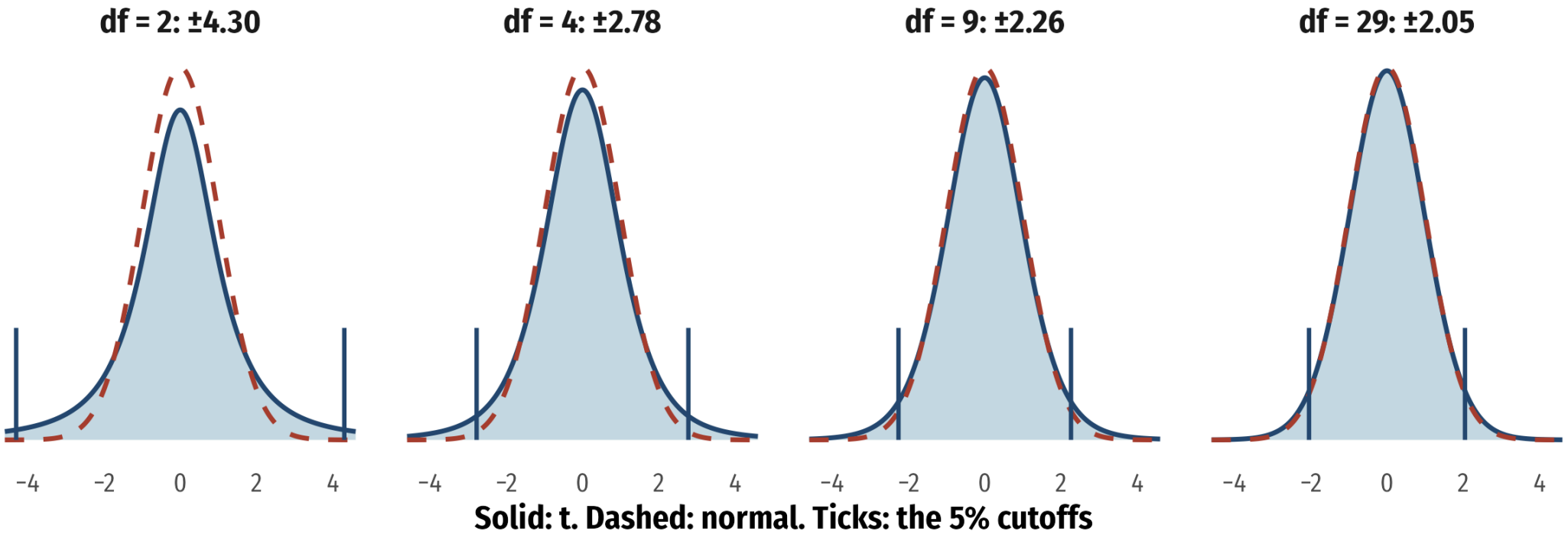
# The t-test

# With n = 5, the pile has fatter tails




Simulator after Kristoffer Magnusson, rpsychologist.com.

# Fewer degrees of freedom, fatter tails




Solid: t. Dashed: normal. Ticks: the 5% cutoffs

A family of curves, one for each df. Near df = 30, it's hard to tell from z

# Each df has its own cutoff

| df | Two tails, α = 0.05 | One tail, α = 0.05 | Two tails, α = 0.01 |
|---|---|---|---|
| 4 | 2.776 | 2.132 | 4.604 |
| 9 | 2.262 | 1.833 | 3.250 |
| 14 | 2.145 | 1.761 | 2.977 |
| **24** | **2.064** | **1.711** | **2.797** |
| 60 | 2.000 | 1.671 | 2.660 |
| 120 | 1.980 | 1.658 | 2.617 |
| Very large (z) | 1.960 | 1.645 | 2.576 |

**In practice, people use t at any n. With large samples it barely differs from z**

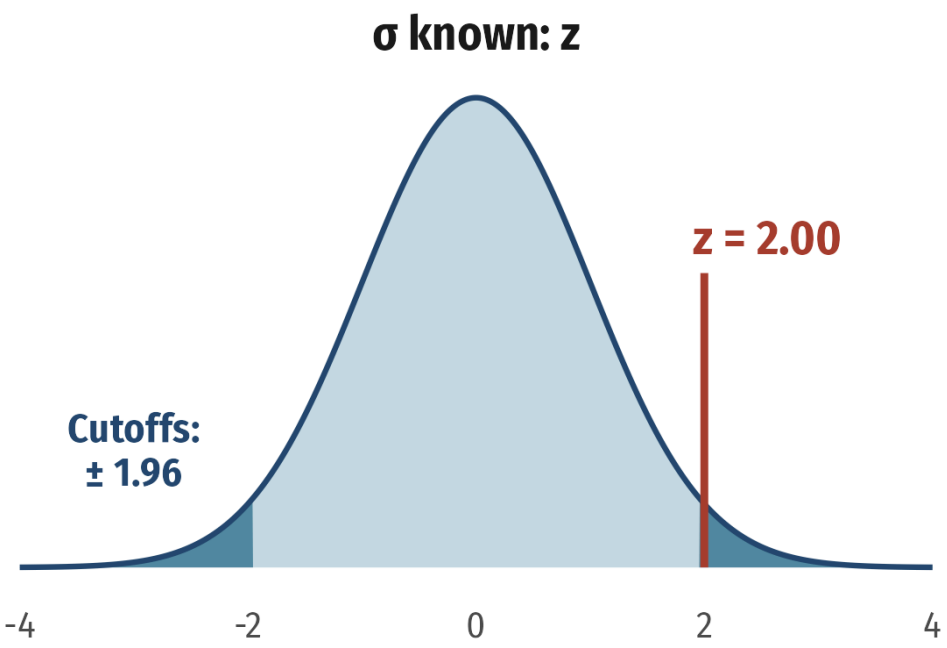
The full t table is in the book's appendix, and comes with the exams

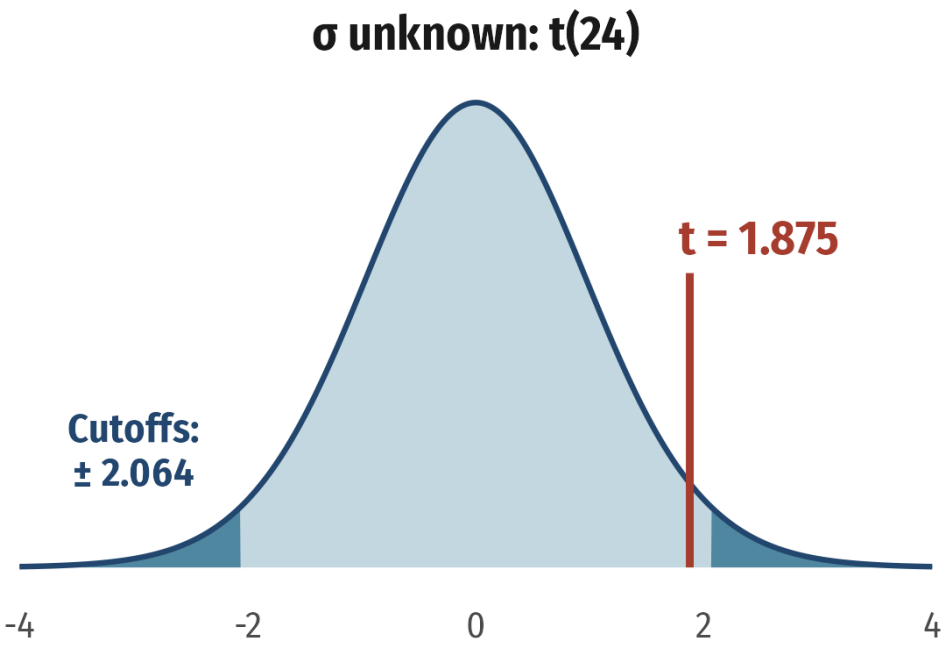
# One example, with and without σ

**n = 25, M = 106, SS = 6,144. $H_0$: μ = 100. α = 0.05, two tails**

| Step | If we know σ = 15 | If we don't |
|---|---|---|
| 1. State $H_0$ | μ = 100 | μ = 100 |
| 2. Set the criteria | z beyond ±1.96 | t(24) beyond ±2.064 |
| 3. Find the spread | $\sigma_M$ = 15 / √25 = 3.00 | $\hat{\sigma}$ = √(6,144 / 24) = 16<br>$\hat{\sigma}_M$ = 16 / √25 = 3.20 |
| 3. Compute the score | z = 6 / 3 = 2.00 | t = 6 / 3.2 = 1.875 |
| 4. Decide | Reject $H_0$: 2.00 > 1.96 | Maintain $H_0$: 1.875 < 2.064 |

# Same sample, different decisions




The score fell: $\hat{\sigma} = 16$ is above 15

The cutoff rose: 2.064 is above 1.96

**We report: $t(24) = 1.88$, $p = 0.07$. We maintain $H_0$**

# Try it: A real group of students

24 Mindspark students take 2 to 3 hours a week of extra math lessons, called tuition in India. At baseline: M = 12.21, SS = 282.0

Test $H_0$: $\mu = 11.12$, the mean of all 619 students. Use $\alpha = 0.05$ and two tails. The cutoff is $t(23) = \pm 2.069$

Find $\hat{\sigma}$, $\hat{\sigma}_M$ and t, then decide. Work alone, compare with a neighbor, then we vote

Data: Muralidharan, Singh, and Ganimian (2019).

# Try it: Answer

$\hat{\sigma} = \sqrt{(282.0 \;/\; 23)} = \mathbf{3.50}$

$\hat{\sigma}_M = 3.50 \;/\; \sqrt{24} = \mathbf{0.715}$

$t = (12.21 - 11.12) \;/\; 0.715 = \mathbf{1.52}$

1.52 is inside ±2.069: **maintain $H_0$**

**We report: t(23) = 1.52, p = 0.14. These 24 students aren't surprisingly far from 11.12**

Data: Muralidharan, Singh, and Ganimian (2019).

# Before Tuesday

**Friday, February 5: Lab 5 is Jeopardy, our midterm review. Bring a calculator**

**Sunday, February 7 at 6 PM: Mastery and Problem Set for Chapters 8 and 9**

**Finish Chapter 9 for Tuesday**

# Session 2: Tuesday, February 9

**Thursday in one line: when we estimate σ, z becomes t, and cutoffs depend on df**

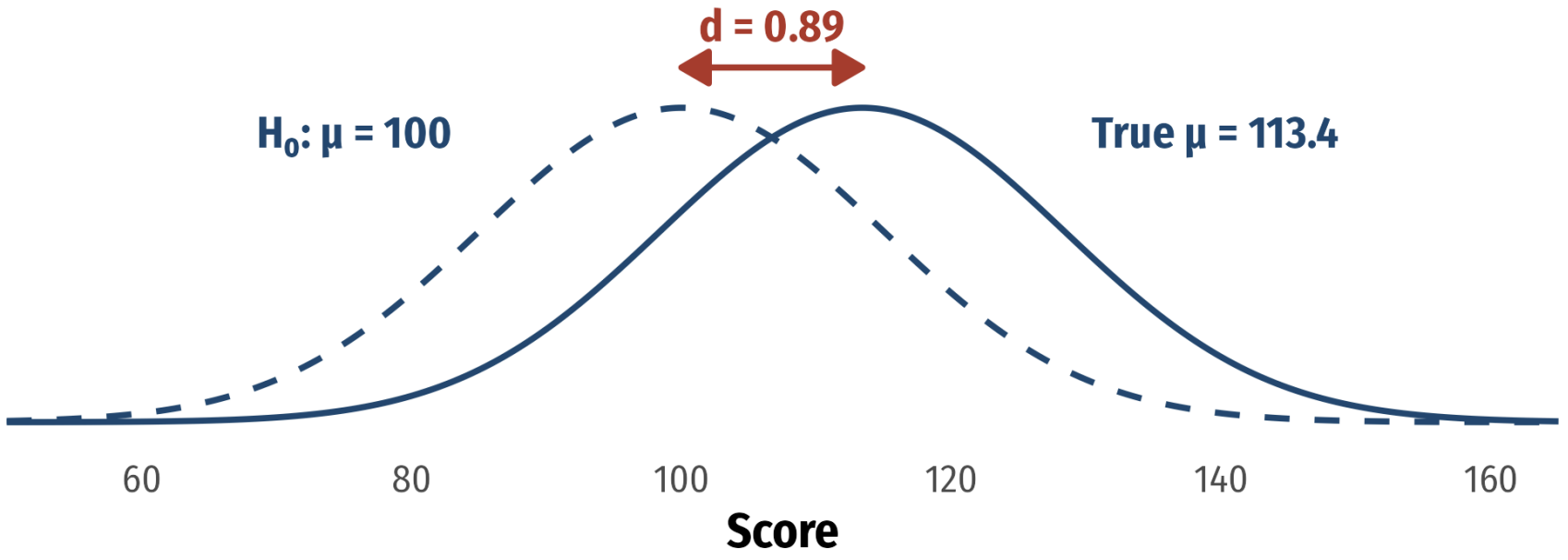
## How big is the effect?

## Confidence intervals

## Intervals and tests

# How big is the effect?

# d says how far apart, in SDs




Cohen's d = (true μ − H₀ μ) / σ = (113.4 − 100) / 15 = 0.89. It's a parameter: it doesn't depend on n

# Hedges' g is the estimate of d

$$d = \frac{\mu_{\text{true}} - \mu}{\sigma} \quad \text{is estimated by} \quad g = \frac{M - \mu}{\hat{\sigma}}$$

**In our example: g = (106 − 100) / 16 = 0.375, so the mean sits about four tenths of an SD above $H_0$**

**Cohen's rough guide: 0.2 is small, 0.5 is medium, and 0.8 is large**

In the book: g is called the estimated Cohen's d. The book also covers $r^2$ here: skip it for now, we come back to it

# Predict it: Four times the people

Respond at **PollEv.com/adb**

**Our example has n = 25, M = 106 and $\hat{\sigma}$ = 16, so t = 1.875 and g = 0.375. Now keep M and $\hat{\sigma}$, and make n = 100. What happens?**

**A** t and g both go up

**B** t goes up, and g stays the same

**C** g goes up, and t stays the same

**D** Neither changes

# Similar effect, different decision

| Group | n | M | g | t | p | Decision |
|---|---|---|---|---|---|---|
| 2 to 3 hours of tuition | 24 | 12.21 | 0.31 | 1.52 | 0.142 | Maintain $H_0$ |
| More than 4 hours | 81 | 12.46 | 0.35 | 3.11 | 0.003 | Reject $H_0$ |

**Both groups sit about a third of an SD above 11.12. Only the larger group is significant**

**Significant means "hard to explain by sampling error". It doesn't mean big**

**These families chose tuition, so neither test shows that tuition works**

Data: Muralidharan, Singh, and Ganimian (2019). $H_0$: $\mu = 11.12$, $\alpha = 0.05$, two tails.

# Confidence intervals

# You already know the margin of error

**A poll: 52% say yes, ± 3 points**

Election polls report one, and charts draw it as a band around the line

It's about two standard errors: how far off this estimate might be

That's the standard error's first job from Topic 6. Now, for a mean

# An interval: estimate ± margin of error

$$M \pm z\,\sigma_M \qquad \text{or} \qquad M \pm t\,\hat{\sigma}_M$$

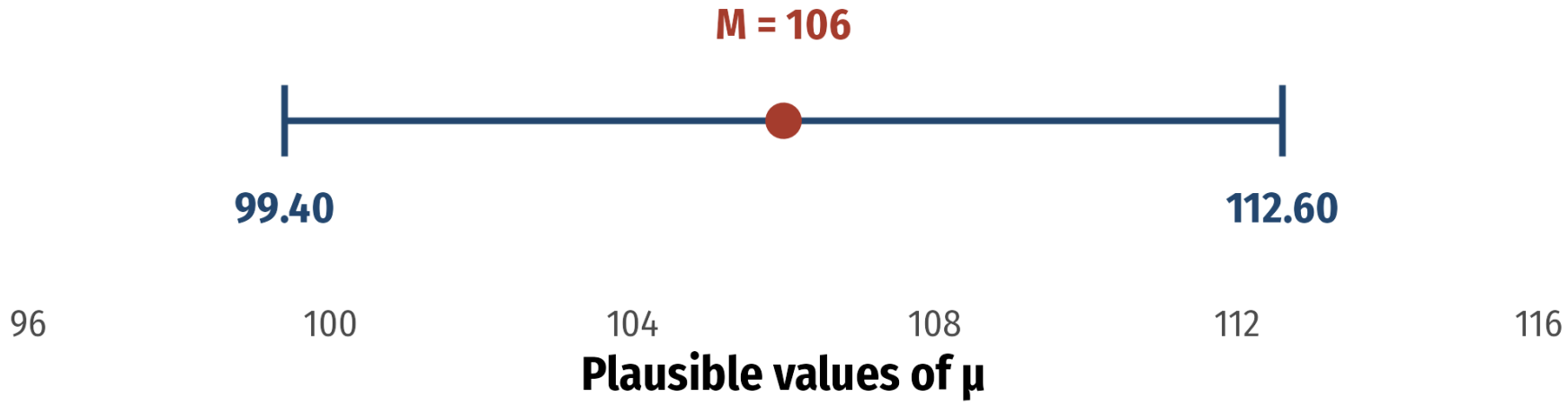
**σ known: z = 1.96 for a 95% interval**

**σ unknown: t from the table, n − 1 df**

**Use the two-tailed 5% cutoff. Margin of error = cutoff × standard error**

In the book: the second one is $M \pm t\ s_M$

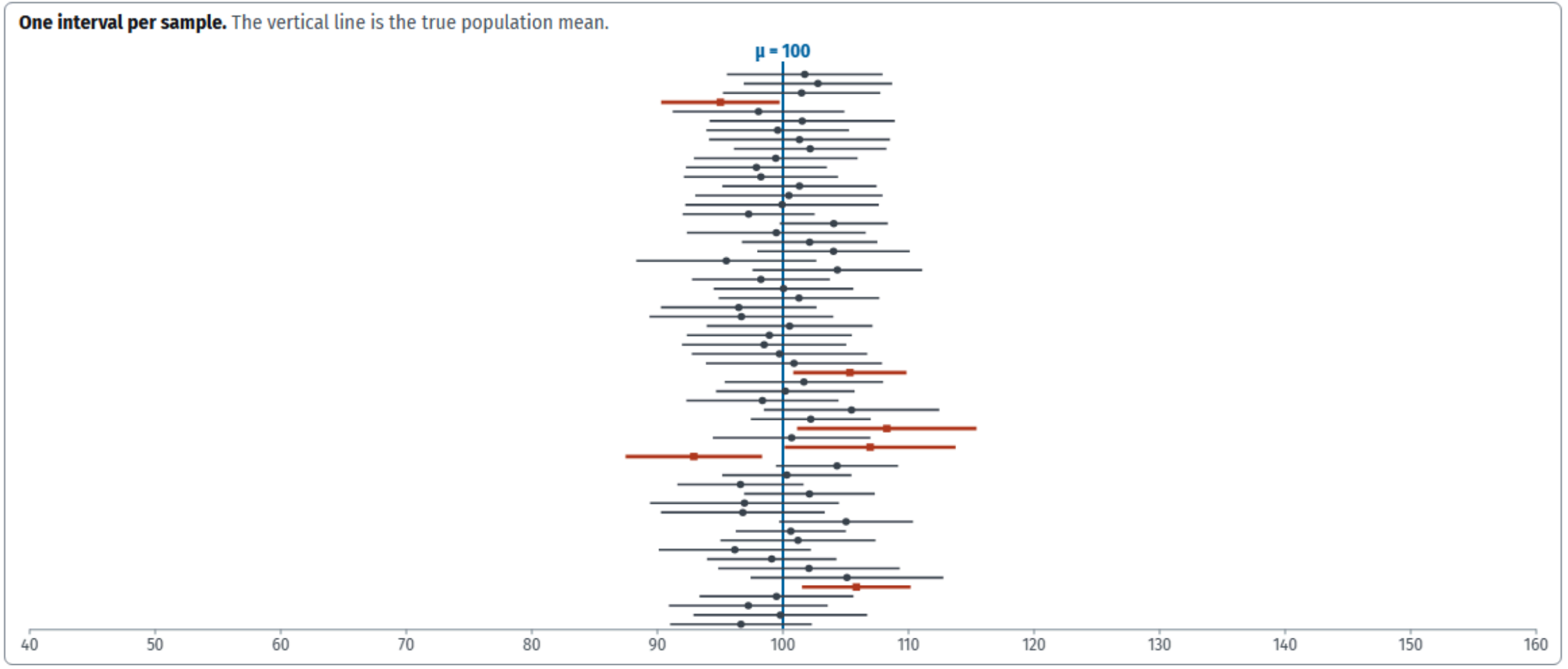
# Our example: from 99.40 to 112.60




$$106 \pm 2.064 \times 3.2 = 106 \pm 6.60$$

**We're 95% confident that μ is between 99.40 and 112.60**

# About 95% of the intervals catch μ




Simulator after Kristoffer Magnusson, rpsychologist.com.

# What the 95% describes

**The method: across many samples, about 95% of the intervals contain μ**

**μ stays put. It's the interval that moves from sample to sample**

**Our one interval either holds μ or it doesn't. We can't tell which**

**On real scores: 94.9% of 5,000 intervals from samples of 25 contain our 619 students' μ = 11.12**

Data: Muralidharan, Singh, and Ganimian (2019).

# Predict it: Ask for more confidence

**We have a 95% interval from 99.40 to 112.60. We want 99% instead, with the same sample**

**What happens to the interval?**

**A** It gets narrower

**B** It gets wider

**C** It stays the same

**D** It shifts to the right

# Predict it: More confidence costs width

| Confidence | n | Cutoff t | Margin of error | Interval |
|---|---|---|---|---|
| 90% | 25 | 1.711 | 5.47 | [100.53, 111.47] |
| 95% | 25 | 2.064 | 6.60 | [99.40, 112.60] |
| 99% | 25 | 2.797 | 8.95 | [97.05, 114.95] |
| 95% | 100 | 1.984 | 3.17 | [102.83, 109.17] |

**The answer is B. To be more sure of catching μ, cast a wider net**

**More people buy the width back: the last row is 95% with n = 100, same M and $\hat{\sigma}$**

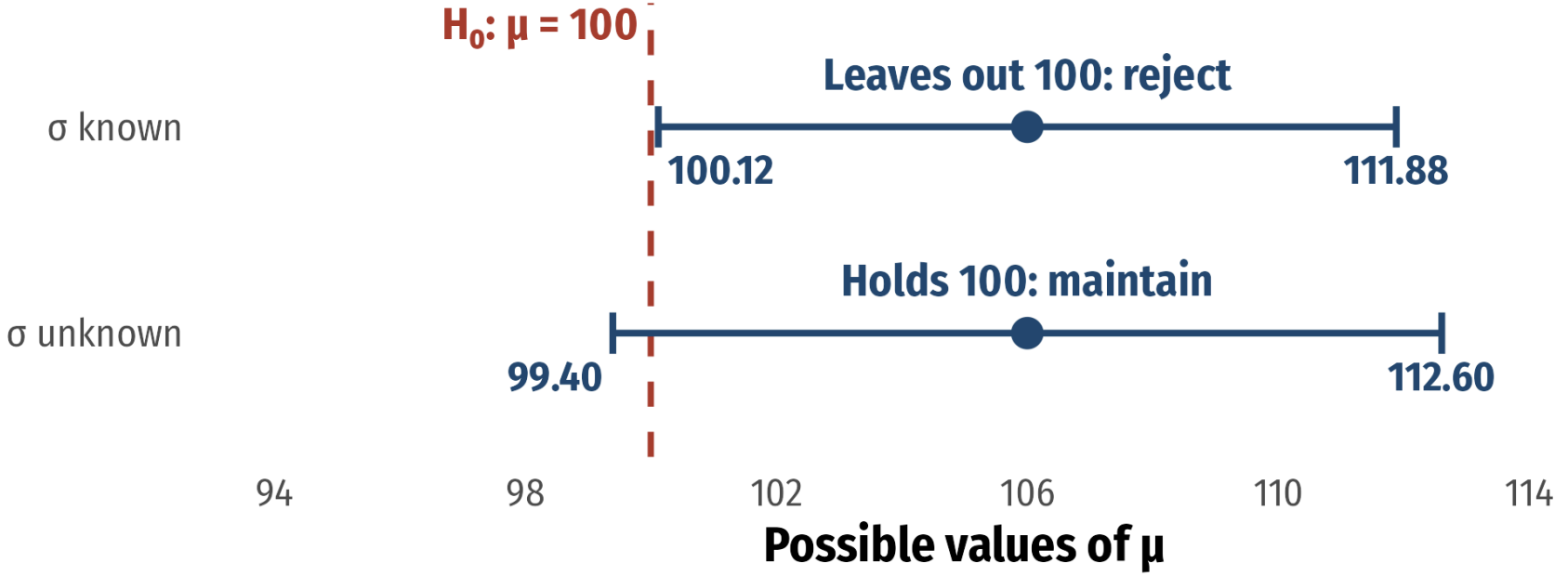
# Intervals and tests

# An interval can run the test for you

| Step | If we know σ = 15 | If we don't |
|---|---|---|
| 1. State $H_0$ | $\mu = 100$ | $\mu = 100$ |
| 2. Set the criteria | 95% interval, $z = 1.96$ | 95% interval, $t(24) = 2.064$ |
| 3. Build the interval | $106 \pm 1.96 \times 3.00$<br>[100.12, 111.88] | $106 \pm 2.064 \times 3.20$<br>[99.40, 112.60] |
| 4. Decide | Reject $H_0$: 100 isn't inside | Maintain $H_0$: 100 is inside |

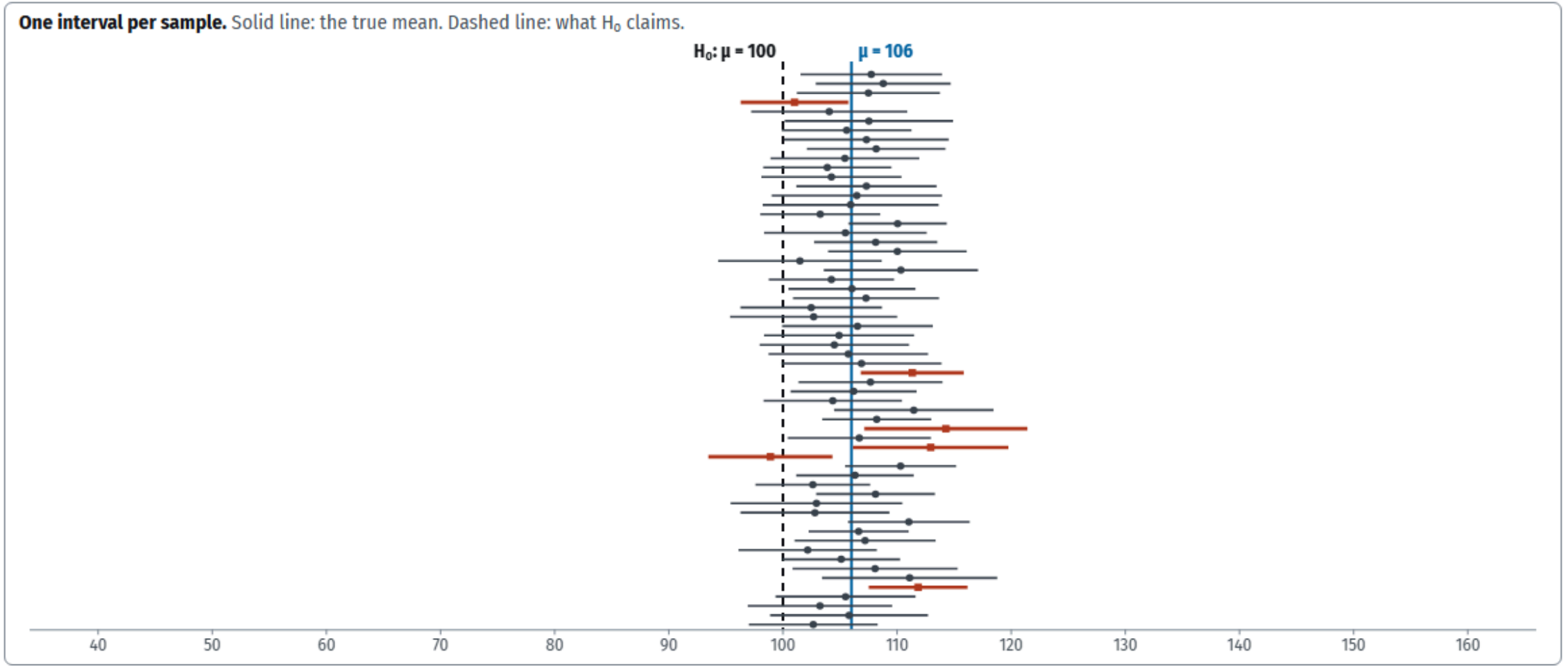
**A 95% interval that leaves out the $H_0$ mean means rejecting $H_0$ at $\alpha = 0.05$, two tails**

# Only one interval leaves out 100




**The t interval is wider, for two reasons: a bigger $\hat{\sigma}$ and a bigger cutoff**

# With a real effect, intervals miss $H_0$




Simulator after Kristoffer Magnusson, rpsychologist.com.

# Try it: The 24 students again

$n = 24$, $M = 12.21$, $\hat{\sigma} = 3.50$, $\hat{\sigma}_M = 0.715$, $H_0$: $\mu = 11.12$, cutoff $t(23) = 2.069$

**Find g, then a 95% confidence interval**

**Does the interval hold 11.12? Does that match Thursday's decision? Work alone, compare with a neighbor, then we vote**

Data: Muralidharan, Singh, and Ganimian (2019).

# Try it: Answer

$g = (12.21 - 11.12) / 3.50 = \mathbf{0.31}$

$12.21 \pm 2.069 \times 0.715 = \mathbf{[10.73, 13.69]}$

**The interval holds 11.12, so we maintain $H_0$, the same decision as Thursday**

**The interval also says what the test didn't: μ could be a bit below 11.12, or as much as 2.6 points above it**

Data: Muralidharan, Singh, and Ganimian (2019).

# One sample, three answers

**The t-test**

Is this mean surprising under $H_0$?

$t(23) = 1.52$

**Effect size**

How big is the gap, in SDs?

$g = 0.31$

**The interval**

Which values of μ fit the data?

$[10.73, 13.69]$

**Report all three. Each one says something the others don't**

Data: Muralidharan, Singh, and Ganimian (2019).

# The blueprint so far

| Blueprint step | Statistic | Bottom of the fraction | In the book |
|---|---|---|---|
| 1. One score | $z$ | $\sigma$ | $\sigma$ |
| 2. One mean, $\sigma$ known | $z$ | $\sigma_M$ | $\sigma_M$ |
| 3a. One mean, $\sigma$ unknown | $t$ | $\hat{\sigma}_M$ | $s_M$ |
| 4. Two means: next | $t$ | Topic 9 | |

**Same three panels: population, one sample, the statistic across samples**

# What you can do now

- Choose one tail or two, and say what each costs
- Say why $\hat{\sigma}$ gives us t, not z, and why df matters
- Run a one-sample t-test by hand, from n, M and SS
- Compute Hedges' g, and tell significance from size
- Build a confidence interval, and use it to test $H_0$

# Thursday's midterm: what to expect

**February 11, in class, 80 minutes**

It covers Chapters 1 to 9, which are Topics 1 to 8

It's free response: show your work and explain your thinking

z and t tables and a formula sheet come with it, in both sets of symbols

Bring a calculator. No other devices

# Next steps

**Thursday, February 11**

The midterm, in class. Nothing is due this week

**Friday, February 12: Lab 6**

One-sample t-tests and confidence intervals. Bring a calculator and a laptop

**For Tuesday, February 16: read Chapters 10 and 11. Topic 9 is t-tests for two samples**